Synthesis: Algorithm II Predictive Propensity — Full Technical Research Synthesis

Algorithm II: Predictive Propensity — Technical Research Synthesis

FirstKnock Sales OS | PropTech Field Sales Platform

PART 1: PREDICTIVE FEATURE ENGINEERING FOR REAL ESTATE PROPENSITY SCORING

1.1 Time-Based Decay Functions

While specific mathematical formulations for exact λ parameters in exponential decay ($f(t) = e^{-\lambda t}$) are not explicitly detailed in the foundational literature, advanced real estate modeling relies heavily on survival analysis to understand ownership duration [1][2].

The **Cox proportional hazards model**provides the primary mathematical framework for examining factors that influence the likelihood and timing of a property sale over time [3].

- **Implementation Guidance:**Models utilizing the Cox framework reveal that property characteristics significantly impact ownership duration; for instance, poor location and larger floor areas decrease the odds of a quick sale [3].
- **Behavioral Adjustments:**Rather than relying solely on static decay functions, scoring pipelines must mathematically account for the disposition effect and loss aversion, which dictate tendencies to hold or sell based on property return and equity [5].
- **Pipeline Integration:**Modern AI tools ingest ownership duration as a core feature alongside billions of other data points to predict property sellability [4].

1.2 Neighborhood Context Enrichment

Effective feature engineering requires grouping raw data into reproducible schemas that capture local quirks, transaction timing, and geographic identifiers [6]. Machine learning models analyze these market condition indicators to predict seller activity [7].

- **Sales Velocity and Absorption Rates:**Absorption rate is a critical engineered feature measuring how quickly inventory clears in specific price bands [8]. It provides a real-time pulse on supply and demand [9].

Market Condition	Absorption Rate	Market Characteristics	Citation
Buyer's Market	< **15%**	Longer property times, softer prices	[9]
Balanced Market	**15%- 20%**	Equilibrium between supply and demand	[9]
Seller's Market	> **20%**	High demand, fast sales, high competition	[9]

- **Price Trend Momentum and DOM:** Days on market (DOM) and list-to-sale ratios are essential pre-appointment data points [8]. Features are engineered from local employment rates, school ratings, new construction permits, and neighborhood development plans to capture market momentum [7]. Propensity scoring systems incorporate these patterns as strong signals for sales momentum [10].

1.3 Composite Motivation Indicators

Seller motivation exists on a continuum, with higher urgency indicating a greater likelihood to accept below-market offers [11]. Predictive analytics identify these sellers before properties hit the market [11].

- **Equity Band Engineering:** An owner's equity position heavily influences selling behavior [12]. Owners with higher Loan-to-Value (LTV) ratios tend to set higher asking prices, experience longer times on the market, and receive higher sale prices [12].
- **Distress Signal Stacking:** AI models analyze financial pressure and distress signals to qualify leads [4][11]. Tools generate a specific "Wholesale Score" indicating the likelihood of a property selling at a discounted price, serving as a proxy for financial distress [13].
- **Life Event Proxies:** Predictive models stack life events such as divorce, job relocation, and demographic shifts to identify owners who genuinely need to sell quickly [11][14].
- **Multi-Signal Composite Scoring:** Advanced propensity scoring predicts the likelihood of a property selling within **6–12** months by analyzing over **800** distinct signals [10][15]. Scores are normalized on a scale from **0** to **1,000** (with **500** as the median), where higher scores indicate a greater propensity to sell within **90** days [13].

1.4 RentCast API Signal Extraction

While specific feature importance rankings for the RentCast API are limited to property type classifications in the provided documentation [16], the broader PropTech landscape highlights high-signal fields:

- **Predictive Fields:** Models rely heavily on financial health, behavioral triggers, historical data, and property condition indicators [4][10][13].

- **Valuation Gaps:**Machine learning models utilize crowdsourced lease comparables and hedonic price indices—incorporating property attributes and financial data—to track market changes accurately [17].
- **Interaction Features:**Systems evaluate the intersection of owner behaviors, market trends, and financial distress to assign comprehensive sellability scores [4].

1.5 Advanced Weighting Frameworks

The industry has moved away from simple rule-based weighted sums toward advanced, non-linear machine learning frameworks [18].

> *"A mortgage broker improved lead scoring by replacing a rule-based system with a predictive model using AI… The project used an XGBoost model trained on structured data to generate probabilistic scores, significantly enhancing lead qualification accuracy."*[18]

- **Gradient Boosted Trees:XGBoost**models are actively deployed to engineer features like behavioral velocity and firmographic signals, outputting highly accurate probabilistic scores [18].
- **Support Vector Machines (SVM):**SVM classification, combined with recursive feature selection, analyzes factors influencing the sale-to-list ratio [19][20].
- **Expected Lift:**By analyzing millions of variables, modern ML models provide property valuations with up to **95%**accuracy [21]. Systems leverage both supervised and unsupervised learning to identify subtle, non-linear signs of selling intent [10][15].

PART 2: DYNAMIC WEIGHT CALIBRATION

2.1 Bayesian Updating Framework

Dynamic weight calibration relies on Bayesian inference to continuously update conversion probabilities. The core is Bayes' Theorem: $P(\theta|data) \propto P(data|\theta) \times P(\theta)$ [22][23].

For binary conversion outcomes (e.g., listing vs. not listing), the **Beta-Binomial model**is utilized [24][25]. The Beta distribution acts as the conjugate prior for the Binomial likelihood.

- **Parameter Update Rules:**$\alpha_{new} = \alpha + conversions$ [24]$\beta_{new} = \beta + non-conversions$ [24]
- **Example:**A prior of Beta(**3**, **17**) with field data of **720**conversions and **3,280**non-conversions updates seamlessly to Beta(**723**, **3297**) [24].
- **Thompson Sampling:**Multi-Armed Bandit (MAB) approaches utilize Thompson Sampling to balance exploration and exploitation [25][26]. Deconfounded Thompson Sampling (DTS) handles nonstationary exogenous factors (like interest rate hikes) using inverse propensity weights [27]. As field reps log funnel progression, the Bayesian MAB updates the

Beta-Bernoulli conjugate priors in real-time [26].

2.2 Market Condition Sensitivity

Models must dynamically adjust feature weights based on macroeconomic environments.

- **Market Regime Detection:**Hidden Markov Models (HMMs) infer unobservable market regimes (hot, cold, neutral) based on absorption rates, DOM, and list-to-sale ratios using transition probabilities and emission distributions [28]. Markov-switching dynamic factor models at the MSA level can identify bust phases approximately **2**years earlier than monthly prices [29].
- **Conditional Adjustments:**House price seasonality has increased, influenced by weather and holidays, requiring updated seasonal adjustments for over **400**metropolitan areas [30][31]. Propensity weights for “spring selling season” must be dampened during winter. MSAs with higher signal-to-noise ratios dictate that urban, suburban, and rural models require distinct baseline weightings [29][31].

2.3 Statistical Validation Methods

Validating models requires specialized techniques due to highly imbalanced data (typical door-to-door conversion rates are **1-5%**). Modern systems analyze over **1 billion**data points to achieve predictive accuracies up to **82.33%**[32].

> ***Holdout Set Design:***_Temporal holdouts (training on past data, testing on future data) are strictly required for real estate models to prevent data leakage and accurately simulate real-world deployment._

Metric	Application in Real Estate Propensity Scoring	Citation
Lift Curves	Evaluates how much better the model identifies conversions compared to random targeting.	[33]
KS Statistic	Measures separation between distributions of positive and negative leads.	[33]
AUC-ROC	Assesses overall ability to rank a true conversion higher than a non-conversion.	[33][34]
Brier Score	Evaluates accuracy of probabilistic predictions matching actual likelihood of sale.	[33]

Calibration Plots	Ensures predicted probabilities align with actual conversion rates across deciles.	[33]
Precision-Recall	Primary metric for highly imbalanced datasets, focusing on positive prediction accuracy.	[33]

2.4 Continuous Refinement Pipeline

Propensity models require robust MLOps pipelines for continuous refinement against external factors like interest rate fluctuations [35].

- **Drift Detection:** The Population Stability Index (PSI) tracks shifts in input feature distributions (Feature Drift) [36]. Algorithms like ADWIN (Adaptive Windowing) and the Page-Hinkley test detect when the relationship between features and the target variable degrades (Concept Drift) [33].
- **MLOps Architecture:** Industry leaders implement pipelines using open-source tools like Evidently and Metaflow to automate anomaly detection [35].
- **Retraining Triggers:** While batch retraining is standard, online learning allows continuous adaptation [33]. Pipelines should utilize performance-based triggers (e.g., AUC drops or severe PSI drift) rather than time-based schedules [35][36].

PART 3: GEOSPATIAL CLUSTERING FOR REVENUE DENSITY

3.1 DBSCAN Parameter Optimization

DBSCAN identifies clusters of arbitrary shapes by detecting dense regions separated by sparse areas [37]. Points are classified as Core, Border, or Noise [37][38].

- **Parameter Selection:** The optimal `eps` (epsilon) is found using a **k-distance graph**, locating the “elbow” where distances sharply increase [40][41]. `MinPts` is typically set to at least twice the dataset dimensionality [40]. Sparse suburban grids require larger `eps` and lower `MinPts`, while dense urban grids need smaller `eps` and higher `MinPts` [41].
- **HDBSCAN Advantages:** HDBSCAN incorporates hierarchical clustering, eliminating the need to manually set `eps` [42][43]. It is highly robust to outliers and handles varying densities in complex PropTech datasets [42][44]. Both algorithms are available in scikit-learn and utilize the **Haversine metric** for accurate latitude/longitude distance calculations [41][42][43].

3.2 K-Means Optimal K Selection

While K-Means struggles with varying densities and outliers [44], it is foundational for baseline territory sizing.

- **Optimal K Selection:** The Elbow Method plots inertia against K. The **Silhouette Score** evaluates cluster distinctness: $s(i) = \frac{b(i) - a(i)}{\max(a(i), b(i))}$ [45][46]. The Gap Statistic compares within-cluster dispersion against a null reference distribution.
- **Scaling:** K-Means++ uses a D^2 weighting algorithm for better centroid initialization. Mini-batch K-Means is essential for processing massive PropTech datasets (e.g., **140M+** property records) in small random batches to drastically reduce computation time.

3.3 Propensity-Weighted Clustering

Integrating propensity scores transforms geographic clustering into revenue-optimized territory design.

- **Spatial Autocorrelation:** Exploratory Spatial Data Analysis (ESDA) assesses if propensity scores follow spatial patterns [47].
- **Hotspot Analysis:** The **Getis-Ord Gi*** statistic identifies statistically significant spatial concentrations (hot and cold spots) of high or low propensity, distinguishing economic activity from random noise [48][49].
- **Implementation:** Python's **PySAL** library is the standard open-source tool for executing these spatial statistical analyses [50][51].

3.4 Revenue Density Maximization

Nearly two-thirds of B2B companies find their territory design ineffective [52].

- **Optimization Models:** Territory design utilizes **p-median** and **capacitated clustering** models, constraining cluster sizes to match a sales rep's daily capacity (e.g., **40-60** houses/day) [52].
- **Contiguous Zone Design:** The **REDCAP** algorithm partitions maps into contiguous zones [52][53]. This is enhanced using an exact dynamic programming approach with **multi-valued decision diagrams (MDDs)** to optimally remove edges from a spatially contiguous tree [53].
- **Graph-Based Frameworks:** Advanced models balance workload equity and minimize unused trips by modeling customers and travel paths as nodes and edges [54].

3.5 Cluster Quality Metrics

Validation techniques are strictly required to ensure balanced territories [45][52].

- **Davies-Bouldin Index:** Measures average similarity ratio; lower is better [45][52].
- **Calinski-Harabasz Score:** Evaluates between-cluster vs. within-cluster dispersion; higher is better [45][52].
- **Dunn Index:** Assesses cluster quality by identifying compact, well-separated clusters [52].
- **Intra-cluster Propensity Variance:** A domain-specific metric evaluating the homogeneity of sales opportunities within a territory to ensure equitable workload and revenue potential [52][53].

4.1 2-Opt Refinement

The 2-opt algorithm is a local search heuristic that optimizes the Traveling Salesman Problem (TSP) by eliminating route crossings [55][56].

- **Mathematical Formulation:** A 2-opt move removes two edges (t1, t2) and (t3, t4) and reconnects the nodes as (t1, t3) and (t2, t4) [55][57]. The swap executes if the new path is strictly shorter.
- **Algorithm Complexity:** For symmetric TSP cases, calculating the change in route length (`lengthDelta`) requires only **O(1)** operations [55].
- **Implementation:** 2-opt is applied to refine initial tours generated by construction heuristics like nearest-neighbor, systematically untangling the route until a local optimum is reached [55][57].

4.2 Loops Not Lines Heuristic

Routing topology significantly impacts efficiency, with a mathematical preference for circular (loop) routes over linear paths.

- **Mathematical Case:** The Christofides algorithm guarantees that for cycles (loops), the tour length will be less than **3/2** (**1.5x**) times the optimal solution [58][60][61]. For linear paths with two specified endpoints, performance degrades to up to **5/3** times the optimal length [61].
- **Cognitive Routing:** Humans naturally rely on hierarchical clustering and Multidimensional Scaling (MDS) rather than strict Euclidean distances when solving TSP in non-Euclidean environments [62][63].

4.3 Virtual Node Injection

The mechanics of virtual node injection for scheduled breaks are modeled using the Vehicle Routing Problem with Time Windows (VRPTW).

- **VRPTW Formulation:** Optimizes routes by ensuring specific nodes are visited within strict time frames, minimizing travel costs while respecting capacity constraints [66].
- **Duration Constraints:** VRPTW with Duration Constraints (VRPTWDC) incorporates flexible departure times and strict route duration limits using an exact branch-cut-and-price algorithm, analogous to enforcing break periods [70].
- **Implementation:** Practical implementation is achieved using solvers like Google OR-Tools [66].

4.4 Real-Time Route Adjustment

Dynamic route optimization requires algorithms that adapt to real-time changes without full recalculation.

- **LENS:** Learning-Enhanced Neighborhood Selection integrates machine learning into Large Neighborhood Search (LNS) to dynamically "destroy and repair" solution parts [67].
- **RL-AVNS:** Reinforcement learning dynamically selects neighborhood operators based on real-time states, guided by a transformer-based neural policy network [68].
- **Time Window Learning:** Multi-armed bandit algorithms learn implicit multiple time windows from historical data to adapt dynamically to contextual factors [65].

4.5 Fatigue Modeling

Optimizing routes for human endurance requires integrating fatigue constraints.

- **Biomathematical Fatigue Constraints (BFCs):** Traditional Hours of Service are mathematically insufficient. Advanced algorithms integrate BFCs into tabu search heuristics to optimize routes specifically for fatigue reduction [64].
- **Workload Balancing:** Multi-objective genetic algorithms utilize multidimensional scaling and fuzzy clustering for preprocessing, leading to travel time reductions of up to **69%**, directly reducing physical exertion [69].

PART 5: INTEGRATED ARCHITECTURE RECOMMENDATIONS

Based on the synthesized research, the following architecture is recommended for the FirstKnock Sales OS:

- **End-to-End Pipeline:** **Data Ingestion:** Extract **140M+** property records via RentCast API, enriching with neighborhood context (absorption rates, DOM) and composite motivation indicators (LTV, Wholesale Score). **Propensity Scoring:** Process features through an XGBoost model to generate probabilistic sellability scores (**0-1000** scale), dynamically calibrated via a Bayesian Beta-Binomial MAB framework. **Geospatial Clustering:** Apply HDBSCAN with Haversine metrics for density-based grouping, followed by REDCAP/MDD algorithms to enforce contiguous, capacitated zones (**40-60** houses/day). Validate with Intra-cluster Propensity Variance. **Routing Optimization:** Generate baseline routes via nearest-neighbor, refine with 2-Opt **O(1)** edge swaps to create loop topologies, and enforce VRPTW constraints for fatigue management.
- **Technology Stack:** Python, Scikit-learn (HDBSCAN, Mini-batch K-Means), PySAL (Getis-Ord Gi*), XGBoost, Google OR-Tools (VRPTW), Evidently/Metaflow (MLOps drift detection).
- **Implementation Priority:** *Quick Wins:* XGBoost baseline scoring, Mini-batch K-Means territory sizing, 2-Opt route refinement. *Long-Term Builds:* Bayesian MAB weight calibration, HDBSCAN + REDCAP contiguous zoning, RL-AVNS real-time route adjustment.
- **Key Risks & Mitigations:** *Risk:* Concept/Feature Drift due to market volatility. *Mitigation:* Implement Evidently MLOps pipelines tracking PSI and triggering retraining on AUC drops. *Risk:* Computational bottleneck on 140M records. *Mitigation:* Utilize Mini-batch K-Means and

O(1) symmetric 2-Opt evaluations.

- **Expected Performance:**Up to **95%**valuation accuracy, **82.33%**predictive conversion accuracy, and up to **69%**reduction in travel time via clustering-based fatigue modeling.

PART 6: KEY REFERENCES & SOURCES

Note: The following citations correspond to the foundational research texts provided in the system context.

Predictive Feature Engineering & Scoring

- [1][2][3]: Survival analysis and Cox proportional hazards models for ownership duration.
- [4][10][11][13][15]: AI propensity scoring, distress signal stacking, Wholesale Scores, and multi-signal composites.
- [5]: Disposition effect and loss aversion mathematics.
- [6][7][8][9]: Neighborhood context, absorption rates, and market momentum.
- [12][14]: Equity band engineering (LTV) and life event proxies.
- [16][17]: RentCast API property classifications and hedonic price indices.
- [18][19][20][21]: XGBoost, SVM, and ML accuracy improvements over rule-based systems.

Dynamic Weight Calibration & Validation

- [22][23][24][25]: Bayes' Theorem and Beta-Binomial conjugate priors.
- [26][27]: Thompson Sampling and Deconfounded Thompson Sampling (DTS).
- [28][29][30][31]: Hidden Markov Models, market regimes, and geographic seasonality.
- [32][33][34]: Statistical validation metrics (AUC-ROC, KS Statistic, Precision-Recall).
- [35][36]: MLOps, PSI feature drift, and continuous refinement pipelines.

Geospatial Clustering

- [37][38][39][40][41]: DBSCAN parameters, k-distance graphs, and grid density.
- [42][43][44]: HDBSCAN advantages and Haversine metric implementation.
- [45][46]: Silhouette Score and cluster quality metrics.
- [47][48][49][50][51]: Spatial autocorrelation, Getis-Ord Gi*, and PySAL.
- [52][53][54]: Capacitated clustering, REDCAP, MDDs, and intra-cluster propensity variance.

TSP & Routing Optimization

- [55][56][57]: 2-Opt algorithm mathematical formulation and O(1) complexity.
- [58][59][60][61]: Christofides algorithm and loop vs. line heuristics.
- [62][63]: Cognitive routing and multidimensional scaling.
- [64][69]: Biomathematical Fatigue Constraints (BFCs) and workload balancing.
- [65][66][67][68][70]: VRPTW, Google OR-Tools, LENS, and RL-AVNS dynamic routing.

References

[1] "Waiting to Be Sold: Prediction of Time-Dependent House Selling Probability". https://ieeexplore.ieee.org/document/7796933/

[2] "Modeling real estate dynamics using survival analysis". https://ieeexplore.ieee.org/document/8750715

[3] "(PDF) Flat location and size as a determinant of homeownership duration". https://www.researchgate.net/publication/362971743_Flat_location_and_size_as_a_determinant_of_homeownership_duration

[4] "Find Motivated Sellers Easier Than Ever Before with PropPulse AI". https://blog.propertyreach.com/real-estate-news/find-motivated-sellers-easier-than-ever-before-with-proppulse-ai

[5] "Does Property Return Affect Seller Behavior? An Empirical Study of China’s Real Estate Market". https://mdpi-res.com/d_attachment/behavsci/behavsci-13-00055/article_deploy/behavsci-13-00055.pdf?version=1673001775

[6] "Zillow Home Value (Zestimate) Prediction in ML: A Practical 2026 Blueprint – TheLinuxCode". https://thelinuxcode.com/zillow-home-value-zestimate-prediction-in-ml-a-practical-2026-blueprint/

[7] "Predicting Home Sales: Machine Learning & AI in Real Estate". https://batchdata.io/blog/likelyhood-to-sell-data-in-real-estate-machine-learning-and-ai

[8] "Data-Driven Listing Presentation: How to Win More Listings With Absorption Rates, Pricing Lanes, and a 30-Day Plan | Medium". https://medium.com/%40srockstad/the-art-of-the-listing-presentation-win-more-listings-with-data-e2fe0df248cf

[9] "What is absorption rate in real estate: How to read market health at a glance | Saleswise". https://www.saleswise.ai/blog/what-is-absorption-rate-in-real-estate

[10] "Propensity Scoring in Real Estate Whitepaper | BatchData Blog". https://batchdata.io/blog/propensity-scoring-real-estate-whitepaper

[11] "Motivated Seller Leads: Definition, Types, and What “Motivation” Really Means in 2026". https://ispeedtolead.com/blog/motivated-seller-leads-2026/

[12] "Equity and Time to Sale in the Real Estate Market". https://econpapers.repec.org/article/aeaaecrev/v_3a87_3ay_3a1997_3ai_3a3_3ap_3a255-69.htm

[13] "What are Sellability Scores? - LeadFlow". https://lfsoftwaresupport.kayako.com/article/1057-what-are-sellability-scores

[14] "Close More Deals: Find the Most Motivated Sellers with Data - Listing3D Insights". https://listing3d.com/insights/close-more-deals-find-the-most-motivated-sellers-with-data/

[15] "Propensity Scoring in Real Estate Whitepaper | BatchData Blog". https://batchdata.io/blog/propensity-scoring-real-estate-whitepaper

[16] "Property Types | RentCast API". https://developers.rentcast.io/reference/property-types

[17] "Data Science: Market Rents on CompStak - CompStak". https://compstak.com/blog/market-rents-on-compstak-whitepaper

[18] "Case Study: How a Broker Used AI to Enhance Lead Scoring - AutomationsLogs". https://automationlogs.com/real-estate/case-study-how-a-broker-used-ai-to-enhance-lead-scoring/

[19] "Machine Learning Insights: Exploring Key Factors Influencing Sale-to-List Ratio—Insights from SVM Classification and Recursive Feature Selection in the US Real Estate Market". https://www.mdpi.com/2075-5309/14/5/1471

[20] "Machine Learning Insights: Exploring Key Factors Influencing Sale-to-List Ratio—Insights from SVM Classification and Recursive Feature Selection in the US Real Estate Market". https://www.mdpi.com/2075-5309/14/5/1471

[21] "Machine Learning in Real Estate Sales: Smarter Pricing & Lead Generation Guide". https://www.articsledge.com/post/machine-learning-real-estate-sales

[22] "Bayesian Inference for A/B Testing: A Practical Guide with Python Examples – Taylor Scott Amarel". https://taylor-amarel.com/2025/05/bayesian-inference-for-a-b-testing-a-practical-guide-with-python-examples/

[23] "Bayesian A/B Testing for Website Conversion Rate Optimization". https://medium.com/@temitopeakinpelu98/bayesian-a-b-testing-for-website-conversion-rate-optimization-e4aeb5dbb1f9

[24] "Bayesian A/B Testing in SaaS Growth: Faster Decisions Without Guesswork – Decision Driven Test Repository!' GrowthLayer.app". https://growthstrategylab.com/bayesian-a-b-testing-in-saas-growth-faster-decisions-without-guesswork/

[25] "grizbayr: Bayesian Inference for A|B and Bandit Marketing Tests". https://rdrr.io/cran/grizbayr

[26] "Optimizing Ad Recommendations Using A Bayesian Multi-Armed Bandit Approach". https://www.itm-conferences.org/articles/itmconf/pdf/2025/09/itmconf_cseit2025_04026.pdf

[27] "[PDF] Adaptivity and Confounding in Multi-Armed Bandit Experiments". https://arxiv.org/pdf/2202.09036

[28] "What State Is the Market In? A First-Principles Guide to HMMs". https://kniyer.substack.com/p/regime-detection-part-1-hidden-markov

[29] "The more, the better? Forecasting gains from high-frequency housing prices in a Markov-switching dynamic factor model | Macroeconomic Dynamics | Cambridge Core". https://www.cambridge.org/core/journals/macroeconomic-dynamics/article/abs/more-the-better-forecasting-gains-from-highfrequency-housing-prices-in-a-markovswitching-dynamic-factor-model/CF78A37B6F096F8582CE9A1D5B80B320

[30] "Applying Seasonal Adjustments to Housing Markets". https://www.huduser.gov/portal/periodicals/cityscape/vol24num3/ch4.pdf

[31] "Working Paper 22-03: Applying Seasonal Adjustments to Housing Markets | FHFA". https://www.fhfa.gov/research/papers/wp2203

[32] "Propensity Scoring in Real Estate Whitepaper | BatchData Blog". https://batchdata.io/blog/propensity-scoring-real-estate-whitepaper

[33] "What Is Propensity Modeling? A Guide to Predicting Customer Behavior". https://batchdata.io/uncategorized/what-is-propensity-modeling

[34] "MLSeries#3: Lead Scoring with Logistic Regression: A Practical Guide to Building Reliable ML Pipelines with Cross-Validation | by gaetan daumas | Medium". https://medium.com/@gaetan.daumas/mlseries-3-lead-scoring-with-logistic-regression-a-practical-guide-to-building-reliable-ml-7d602006934c

[35] "How Realtor.com Adopted In-house Feature Drift Tooling Built on Open Source - realtor.com Tech Blog". https://techblog.realtor.com/how-realtor-com-adopted-in-house-feature-drift-tooling-built-on-open-source/

[36] "How to Implement Model Drift Detection". https://oneuptime.com/blog/post/2026-01-25-model-drift-detection/view

[37] "DBSCAN and HDBSCAN: The Complete Guide to Density-Based Clustering - perfecXion.ai". https://perfecxion.ai/articles/dbscan-hdbscan-complete-guide.html

[38] "Mastering DBSCAN with Scikit-learn: A Practical Guide - codepointtech.com". https://codepointtech.com/mastering-dbscan-with-scikit-learn-a-practical-guide/

[39] "DBSCAN — an Easy Clustering Algorithm and also how to optimize it Using grid search". https://medium.com/@revag2014/dbscan-an-easy-clustering-algorithm-and-also-how-to-optimize-it-using-grid-search-69a382b63e85

[40] "How to Choose Optimal Hyperparameters for DBSCAN - stataiml". https://stataiml.com/posts/how_to_set_dbscan_paramter/

[41] "DBSCAN Clustering in ML: A Practical Density-Based Playbook". https://thelinuxcode.com/dbscan-clustering-in-ml-a-practical-density-based-playbook/

[42] "HDBSCAN Clustering: Complete Guide to Hierarchical Density-Based Clustering with Automatic Cluster Selection - Interactive | Michael Brenndoerfer | Michael Brenndoerfer". https://mbrenndoerfer.com/writing/hdbscan-hierarchical-density-based-clustering-automatic-cluster-selection

[43] "DBSCAN vs. HDBSCAN. Which One You Should Use? - stataiml". https://stataiml.com/posts/44_dbscan_vs_hdbscan/

[44] "Advanced Customer Segmentation: A Comprehensive Comparison of HDBSCAN, DBSCAN, and K-Means | by Ruma Sinha | Towards AI". https://pub.towardsai.net/advanced-customer-segmentation-a-comprehensive-comparison-of-hdbscan-dbscan-and-k-means-ce3bcaa7f1a2

[45] "From Math to Marketing: Validating Clusters and Extracting Actionable Insights | by Mariangela Bonghi | Feb, 2026 | Towards AI". https://pub.towardsai.net/from-math-to-marketing-validating-clusters-and-extracting-actionable-insights-264d4443ee3b

[46] "How To Tune HDBSCAN - TDS Archive - Medium". https://medium.com/data-science/tuning-with-hdbscan-149865ac2970

[47] "Exploratory Spatial Data Analysis (ESDA): Spatial Autocorrelation". https://kazumatsuda.medium.com/exploratory-spatial-data-analysis-esda-spatial-autocorrelation-71b5782c19d6

[48] "Python Hotspot Analysis: Identifying Statistically Significant Spatial Clusters". https://medium.com/@stacyfuende/python-hotspot-analysis-identifying-statistically-significant-spatial-clusters-d7dec9f8c23f

[49] "Spatial Statistics and Geostatistics | SAGE India".
https://in.sagepub.com/en-in/ind/spatial-statistics-and-geostatistics/book236769

[50] "Using Python PySAL for Geographically Weighted Regression".
https://medium.com/data-science-collective/using-python-pysal-for-geographically-weighted-regression-fd107df78c98

[51] "PD-31 - PySAL and Spatial Statistics Libraries".
https://gistbok.ucgis.org/bok-topics/pysal-and-spatial-statistics-libraries

[52] "Field Territory Design: The Revenue Leader's Playbook for Balanced, High-Performing Territories - RevenueTools Blog | RevenueTools".
https://www.revenuetools.io/blog/field-territory-design

[53] "Partitioning a Map into Homogeneous Contiguous Regions: A Branch-And-Bound Approach Using Decision Diagrams (Short Paper)".
https://drops.dagstuhl.de/entities/document/10.4230/LIPIcs.CP.2023.45

[54] "Territory Planning Algorithms: Graph-Based Sales Coverage Optimization | International journal of data science and machine learning".
https://inlibrary.uz/index.php/ijdsml/article/view/112006

[55] "2-opt".
https://en.wikipedia.org/wiki/2-opt

[56] "How to Solve the Traveling Salesman Problem - A Comparative Analysis | Tow
https://towardsdatascience.com/how-to-solve-the-traveling-salesman-problem-a-comparative-analysis-39056a916c9f

[57] "Aspects of OR - Traveling Salesperson Problem - Heuristics (Part 2) | Homepage".
https://www.till-heller.de/blog/aspects-of-or-tsp-2

[58] "Christofides Algorithm: The Secret Weapon for Route Optimization".
https://priyadarshanghosh26.medium.com/christofides-algorithm-the-secret-weapon-for-route-optimization-d2b9ec68d66e

[59] "Heuristic Methods for Solving the Traveling Salesman Problem (TSP): A Comparative Study".
https://ieeexplore.ieee.org/document/10293957/

[60] "Travelling Salesman Problem (TSP): Algorithms and Approaches- A Comprehensive Survey and Analysis".
https://zenodo.org/records/15825858

[61] "Analysis of Christofides' heuristic: Some paths are more difficult than cycles".
https://www.sciencedirect.com/science/article/abs/pii/016763779190016I

[62] "Traveling Salesperson Problem with Simple Obstacles: The Role of Multidimensional Scaling and the Role of Clustering".
https://link.springer.com/content/pdf/10.1007/s42113-022-00155-0.pdf?error=cookies_not_supported&code=b64d50cc-9a58-455d-947c-42e239e54ed3

[63] "Cognitive Approaches to the Traveling Salesperson Problem: Perceptual Complexity that Produces Computational Simplicity - AAAI".
https://aaai.org/papers/0005-ss06-02-005-cognitive-approaches-to-the-traveling-salesperson-problem-perceptual-complexity-that-produces-computational-simplicity/

[64] "Long-Haul Vehicle Routing and Scheduling with Biomathematical Fatigue Constraints". https://ideas.repec.org/a/inm/ortrsc/v56y2022i2p404-435.html

[65] "Learning implicit multiple time windows in the Traveling Salesman Problem". https://hal.science/hal-04571639v1/file/Learning_implicit_time_windows_in_the_traveling_salesman_problem__HAL_.pdf

[66] "Vehicle Routing Problem with Time Windows: Complete Guide to VRPTW Optimization with OR-Tools - Interactive | Michael Brenndoerfer | Michael Brenndoerfer". https://mbrenndoerfer.com/writing/vehicle-routing-problem-time-windows-vrptw-optimization-guide

[67] "Learning-Enhanced Neighborhood Selection for the Vehicle Routing Problem with Time Windows". https://arxiv.org/html/2403.08839v1

[68] "Learning to Search for Vehicle Routing with Multiple Time Windows". https://arxiv.org/pdf/2505.23098

[69] "Genetic Algorithm Optimization of Sales Routes with Time and Workload Objectives". https://www.mdpi.com/2673-9909/5/3/103

[70] "A Branch-Cut-and-Price algorithm for the vehicle routing problem with time windows and duration constraints". https://hal.science/hal-05303852v1/document